

FINAL ASSESMENT

SNOWFLAKE + POWERBI

AIM:

Healthcare Appointment No-Show Analytics using Snowflake

Business Scenario

A hospital network wants to analyze patient appointment behavior.

The hospital wants to know:

How many appointments are booked?

How many patients miss their appointments?

Which age group has the highest no-show rate?

Does SMS reminder reduce no-show?

Which neighborhood has more missed appointments?

How can Snowflake automatically load new appointment files?

How can Power BI show the final dashboard?

You are asked to build a Snowflake data pipeline from raw CSV data to a clean reporting table.

PART 1:

Create Snowflake Objects

Create:

Warehouse

Database

Schema

File format

Internal stage

```

-----part a-----
---warehouse
create or replace warehouse healthcare_wh
with
warehouse_size = 'XSMALL'
initially_suspended = TRUE
auto_suspend = 60
auto_resume = TRUE
---database
create or replace database healthcare_db;
-----schema
create or replace schema appointment_shema;
-----file format
create or replace file format csv_format
TYPE = CSV
FIELD_DELIMITER = ','
SKIP_HEADER = 1;
---internal stage
create or replace stage healthcare_stage
file_format = csv_format

```

PART 2:

Scale Warehouse Up and Down

Perform the following:

Create warehouse as XSMALL

Scale it up to SMALL

Scale it down again to XSMALL

Suspend the warehouse manually

```

-----part b-----
---scale warehouse up---
alter warehouse healthcare_wh set warehouse_size='SMALL';
---scale down---
alter warehouse healthcare_wh set warehouse_size='XSMALL';
---suspend warehouse--
alter warehouse healthcare_wh suspend;
---resume warehouse-----
alter warehouse healthcare_wh resume;

```

PART 3:

Download Dataset from Open URL

Use this public CSV dataset:

<https://raw.githubusercontent.com/kb22/Medical-Appointment-No-Show-Prediction/master/data/dataset.csv>

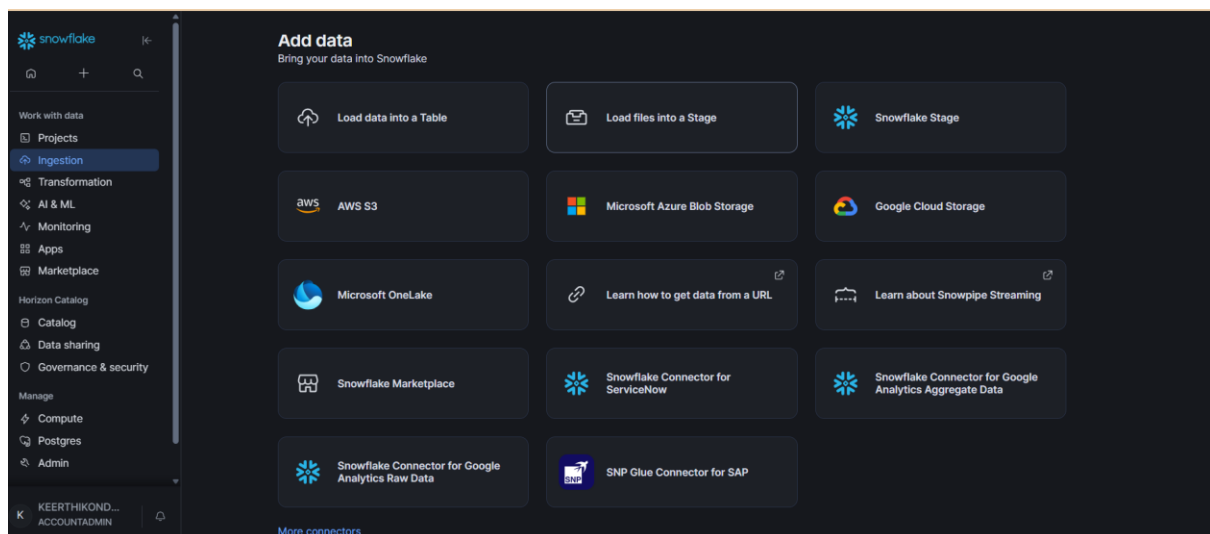
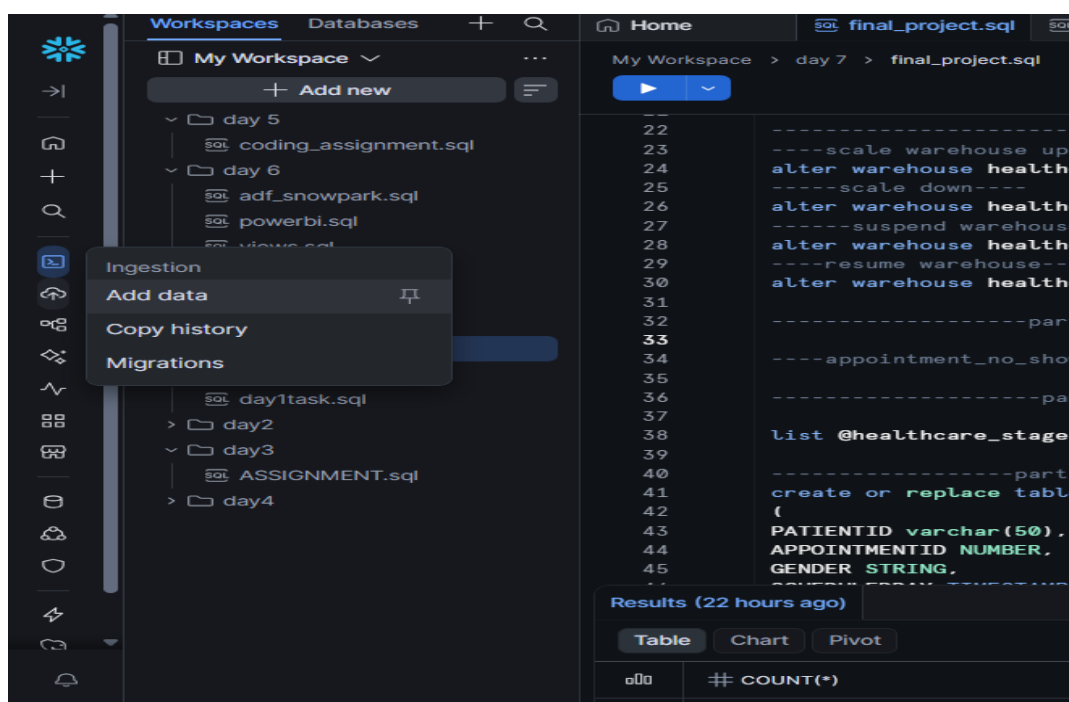
Download the CSV file to your local system.

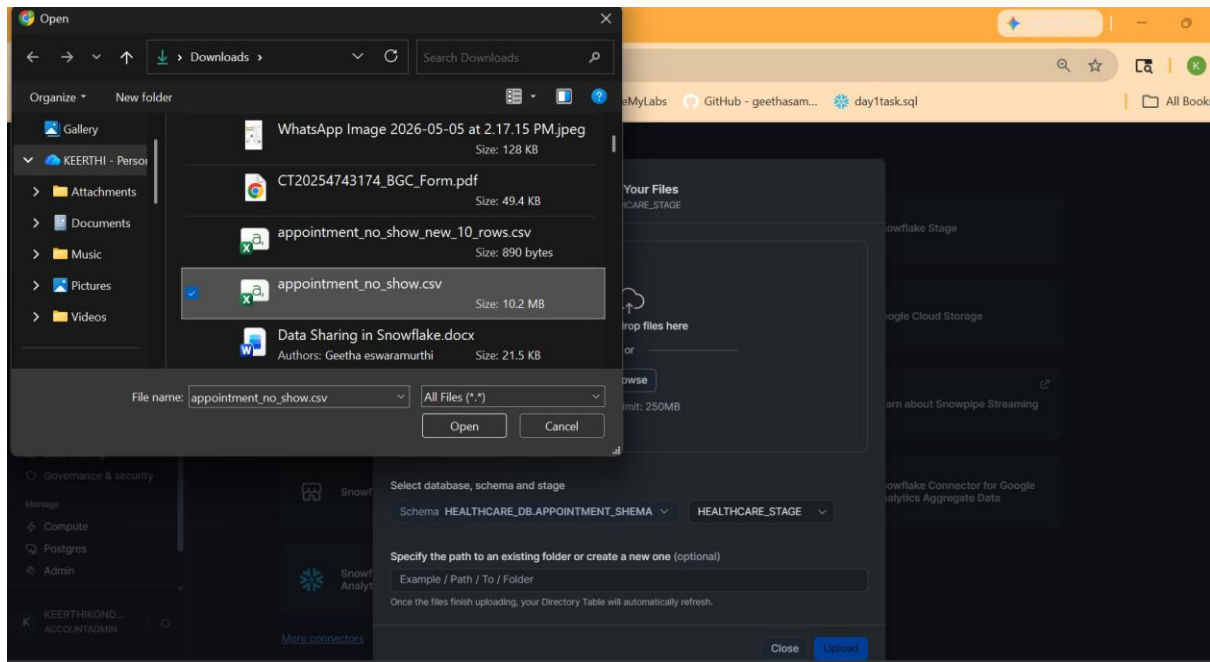
File name

appointment_no_show.csv

Upload CSV into Snowflake Stage

Upload the CSV file into Snowflake internal stage using Snowsight.





PART 4:

LIST FILES IN THE STAGE(INTERNAL)

```

36 -----part d-----
37
38 list @healthcare_stage;
39

```

Results (just now)

#	name	size	md5	last_modified
1	healthcare_stage/appointment_no_show.csv	10739536	724bb5614b9a8f7bf9f67c9ecf2b7d9d	Sat, 23 May 2026 05:31:59 GMT

PART 5:

Create Raw Table

Create a raw table to store CSV data exactly as received.

```

39
40 -----part e-----
41
42 create or replace table appointment_raw
43 (
44     PATIENTID varchar(50),
45     APPOINTMENTID NUMBER,
46     GENDER STRING,
47     SCHEDULEDDAY TIMESTAMP,
48     APPOINTMENTDAY TIMESTAMP,
49     AGE NUMBER,
50     NEIGHBOURHOOD STRING,
51     SCHOLARSHIP NUMBER,
52     HIPERTENSION NUMBER,
53     DIABETES NUMBER,
54     ALCOHOLISM NUMBER,
55     HANDCAP NUMBER,
56     SMS_RECEIVED NUMBER,
57     NO_SHOW STRING
58 );

```

Load Data from Stage to Raw Table

Snowflake COPY INTO <table> loads staged files into a table using the selected file format



Create a transformed final table:

```
-----create final table-----

create or replace table appointment_final
(
Patient_ID varchar(50),
Appointment_ID number,
Gender string,
Age number,
Age_group string,
Scheduled_date timestamp,
Appointment_date date,
Waiting_days number,
Neighborhood string,
Scholarship string,
Hypertension string,
Diabetes string,
Alcoholism string,
Handicap string,
SMS_received string,
No_show_status string,
Attendance_status string
);
```

PART 8:

Use Snowpark Python

Create a Snowpark Python stored procedure to transform data from raw table to final table.

Snowpark Python procedures can run Python logic inside Snowflake and perform queries, updates, and transformations on Snowflake tables.

```
-----snowpark python-----
CREATE OR REPLACE PROCEDURE LOAD_APPOINTMENT_FINAL ()
RETURNS STRING
LANGUAGE PYTHON
RUNTIME_VERSION = '3.9'
PACKAGES = ('snowflake-snowpark-python')
HANDLER = 'main'
AS
$$

from snowflake.snowpark import Session
from snowflake.snowpark.functions import *

def main(session: Session):

    df = session.table("APPOINTMENT_RAW")

    transformed_df = df.select(

        col("PATIENTID").alias("Patient_ID"),

        col("APPOINTMENTID").alias("Appointment_ID"),

        col("GENDER").alias("Gender"),

        col("AGE").alias("Age"),

        when(col("AGE") < 18, "Children")
        .when((col("AGE") >= 18) & (col("AGE") <= 30), "Youth")
        .when((col("AGE") > 30) & (col("AGE") < 60), "Adults")
        .otherwise("Senior Citizens")
```

```
2         .alias("Age_group"),
3
4         col("SCHEDULEDDAY").alias("Scheduled_date"),
5
6         to_date(col("APPOINTMENTDAY")).alias("Appointment_date"),
7
8         datediff(
9             "day",
10            to_date(col("SCHEDULEDDAY")),
11            to_date(col("APPOINTMENTDAY"))
12        ).alias("Waiting_days"),
13
14        col("NEIGHBOURHOOD").alias("Neighborhood"),
15
16        when(col("SCHOLARSHIP") == 1, "Yes")
17        .otherwise("No")
18        .alias("Scholarship"),
19
20        when(col("HIPERTENSION") == 1, "Yes")
21        .otherwise("No")
22        .alias("Hypertension"),
23
24        when(col("DIABETES") == 1, "Yes")
25        .otherwise("No")
26        .alias("Diabetes"),
27
28        when(col("ALCOHOLISM") == 1, "Yes")
29        .otherwise("No")
30        .alias("Alcoholism"),
31
32        when(col("HANDCAP") >= 1, "Yes")
33        .otherwise("No")
34        .alias("Handicap"),
35    )
```

```

        .otherwise("No")
        .alias("Handicap"),

        when(col("SMS_RECEIVED") == 1, "Yes")
        .otherwise("No")
        .alias("SMS_received"),

        col("NO_SHOW").alias("No_show_status"),

        when(col("NO_SHOW") == "Yes", "Missed")
        .otherwise("Visited")
        .alias("Attendance_status")

    )

    transformed_df.write.mode("overwrite").save_as_table("APPOINTMENT_FINAL")

    return "Data Loaded Successfully"
}

--SS:

call LOAD_APPOINTMENT_FINAL();

select * from appointment_final limit 10;

```

176 | select * from appointment_final limit 10;
177

Results (just now)

Table Chart Pivot

	PATIENT_ID	APPOINTMENT_ID	GENDER	AGE	AGE_GROUP	SCHEDULED_DATE	APPOINTMENT_DATE	WAITING_DAYS	NEIGHBORHOOD	SCHOLARSHIP	HYPERTENSION
	29872499...	10.0%									
	34498333...	10.0%									
	+8 more										
1	29872499824296	5642903	F	62	Senior Citizens	2016-04-29 18:38:08.000	2016-04-29	0	JARDIM DA PENHA	No	No
2	558997776694438	5642503	M	56	Adults	2016-04-29 16:08:27.000	2016-04-29	0	JARDIM DA PENHA	No	No
3	4262962299951	5642549	F	62	Senior Citizens	2016-04-29 16:19:04.000	2016-04-29	0	MATA DA PRAIA	No	No
4	867951213174	5642828	F	8	Children	2016-04-29 17:29:31.000	2016-04-29	0	PONTAL DE CAMBURI	No	No
5	8841186448183	5642494	F	56	Adults	2016-04-29 16:07:23.000	2016-04-29	0	JARDIM DA PENHA	No	Yes
6	95985133231274	5628772	F	76	Senior Citizens	2016-04-27 08:36:51.000	2016-04-29	2	REPÚBLICA	No	Yes
7	733688164476661	5630279	F	23	Youth	2016-04-27 15:05:12.000	2016-04-29	2	GOIABERAS	No	No
8	3449833394123	5630575	F	39	Adults	2016-04-27 15:39:58.000	2016-04-29	2	GOIABERAS	No	No
9	56394729949972	5638447	F	21	Youth	2016-04-29 08:02:16.000	2016-04-29	0	ANDORINHAS	No	No
10	78124564369297	5629123	F	19	Youth	2016-04-27 12:48:25.000	2016-04-29	2	CONQUISTA	No	No

PART 9:

Create a stream on the raw table to track newly inserted rows.

A Snowflake stream tracks row-level changes in a source table between transactional points, which is useful for change data capture.

```

-----part 9-----
-----schema creation
create or replace stream hospital_stream
on table appointment_raw;

select * from hospital_stream;

```

```
183 | select * from hospital_stream;
```

```
184
```

Results (just now)

Table Chart Pivot

PATIENTID	APPOINTMENTID	GENDER	SCHEDULEDDAY	APPOINTMENTDAY	AGE	NEIGHBOURHOOD	SCHOLARSHIP	HIPERTENSION	DIABETES
 Query produced no results									

PART 10:

Use Stream for Incremental Loading

Insert new records into the raw table.

Then use the stream to load only new records into the final table.

```
-----part 10-----
-----insert new records into raw table

insert into appointment_raw values
(
  'P10019999E',
  10001,
  'F',
  '2024-05-01 10:00:00',
  '2024-05-05',
  25,
  'CENTRAL',
  1,
  0,
  0,
  0,
  0,
  1,
  'YES'
),
('P10018888E',10003,'F','2024-06-01 10:00:00','2024-06-05',26,'CALIFORNIA',1,0,0,
0,
0,
1,
'YES'
);
```

```
183 | select * from hospital_stream;
```

```
184
```

Results (just now)

Table Chart Pivot

#	SCHOLARSHIP	HIPERTENSION	DIABETES	ALCOHOLISM	HANDCAP	SMS_RECEIVED	NO_SHOW	METADATA\$	METADATA\$ISUPDATE	METADATA\$ROW_ID
1	1	0	0	0	0	1	YES	INSERT	FALSE	f1e846db94f0044f19e2786655ea824a9c00c3c
2	1	0	0	0	0	1	YES	INSERT	FALSE	c39c7b62323fda29fa201d8e0b2dc08b54da2b0

PART 11:

```
-----INSERT STREAM DATA TO FINAL TABLE-----

INSERT INTO APPOINTMENT_FINAL
SELECT

PATIENTID AS Patient_ID,

APPOINTMENTID AS Appointment_ID,

GENDER AS Gender,

AGE AS Age,

CASE
  WHEN AGE < 18 THEN 'Children'
  WHEN AGE BETWEEN 18 AND 30 THEN 'Youth'
  WHEN AGE > 30 AND AGE < 60 THEN 'Adults'
  ELSE 'Senior Citizens'
END AS Age_group,

SCHEDULEDDAY AS Scheduled_date,

TO_DATE(APPOINTMENTDAY) AS Appointment_date,

DATEDIFF(
  DAY,
  TO_DATE(SCHEDULEDDAY),
```



```

TO_DATE(APPOINTMENTDAY)
) AS Waiting_days,

NEIGHBOURHOOD AS Neighborhood,

CASE WHEN SCHOLARSHIP = 1 THEN 'Yes' ELSE 'No' END AS Scholarship,

CASE WHEN HIPERTENSION = 1 THEN 'Yes' ELSE 'No' END AS Hypertension,

CASE WHEN DIABETES = 1 THEN 'Yes' ELSE 'No' END AS Diabetes,

CASE WHEN ALCOHOLISM = 1 THEN 'Yes' ELSE 'No' END AS Alcoholism,

CASE WHEN HANDCAP >= 1 THEN 'Yes' ELSE 'No' END AS Handicap,

CASE WHEN SMS_RECEIVED = 1 THEN 'Yes' ELSE 'No' END AS SMS_received,

NO_SHOW AS No_show_status,

CASE
WHEN NO_SHOW = 'Yes' THEN 'Missed'
ELSE 'Visited'
END AS Attendance_status

FROM hospital_stream
WHERE METADATA$ACTION = 'INSERT';

```

183 | select * from hospital_stream;
184

Results (just now)

Table Chart Pivot 0 rows

PATIENTID	APPOINTMENTID	GENDER	SCHEDULEDDAY	APPOINTMENTDAY	AGE	NEIGHBOURHOOD	SCHOLARSHIP	HIPERTENSION	DIABETES
 Query produced no results									

PART 12:

Create Snowpipe

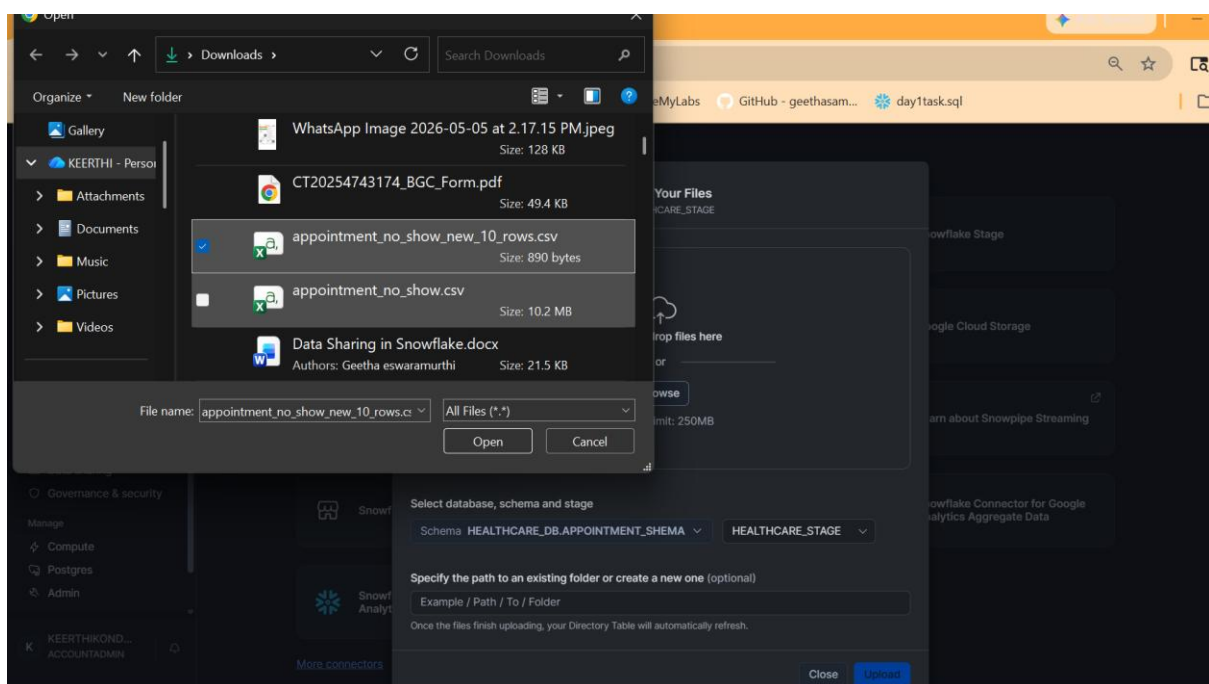
Create a Snowpipe pipe to load new CSV files from the stage into the raw table.

Snowpipe loads files as soon as they are available in a stage, using a pipe object that contains a COPY INTO statement.

```
-----SNOWPIPE--
| Ctrl+I to generate
CREATE OR REPLACE PIPE HOSPITAL_PIPE
AUTO_INGEST = FALSE
AS
COPY INTO APPOINTMENT_RAW
FROM @HEALTHCARE_STAGE
FILE_FORMAT = CSV_FORMAT;

ALTER PIPE HOSPITAL_PIPE REFRESH;

SELECT SYSTEM$PIPE_STATUS('HOSPITAL_PIPE');
```



278
279 | SELECT SYSTEM\$PIPE_STATUS('HOSPITAL_PIPE');
788

Results (just now)

	SYSTEM\$PIPE_STATUS(HOSPITAL_PIPE)
1	{"executionState":"RUNNING","pendingFileCount":0,"lastingestedTimestamp":"2026-05-24T09:32:58.219Z","lastingestedFilePath":"appointment_no_show_new_10_rows.csv","pendingHistoryRefreshJobsCount":0}

FIRST THE COUNT WAS 110529 AFTER PIPE 110539

310 | select count(*) from appointment_raw;

Results (just now)

	COUNT(*)
1	110539

320 | select count(*) from appointment_final;

Results (just now)

	COUNT(*)
1	110539

PART 13:

```
-----revenue view-----

CREATE OR REPLACE VIEW VW_APPOINTMENT_DASHBOARD
AS
SELECT
Patient_ID,
Appointment_ID,
Gender,
Age,
Age_group,
Scheduled_date,
Appointment_date,
Waiting_days,
Neighborhood,
Scholarship,
Hypertension,
Diabetes,
Alcoholism,
Handicap,
SMS_received,
No_show_status,
Attendance_status

FROM APPOINTMENT_FINAL;

select * from VW_APPOINTMENT_DASHBOARD limit 10;
```

	PATIENT_ID	APPOINTMENT_ID	GENDER	AGE	AGE_GROUP	SCHEDULED_DATE	APPOINTMENT_DATE	WAITING_DAYS	NEIGHBORHOOD	SCHOLARSHIP	HYPERTENSION
	29672499...	5626772	F	62	Senior Citizens	4/27/2016	4/29/2016	0	JARDIM DA PENHA	No	No
1	558997776684438	5642503	M	56	Adults	2016-04-29 16:08:27.000	2016-04-29	0	JARDIM DA PENHA	No	No
2	4262962299951	5642549	F	62	Senior Citizens	2016-04-29 16:19:04.000	2016-04-29	0	MATA DA PRAIA	No	No
3	867951213174	5642828	F	8	Children	2016-04-29 17:29:31.000	2016-04-29	0	PONTAL DE CAMBURI	No	No
4	8841166448183	5642494	F	56	Adults	2016-04-29 16:07:23.000	2016-04-29	0	JARDIM DA PENHA	No	Yes
5	95985133231274	5626772	F	76	Senior Citizens	2016-04-27 08:36:51.000	2016-04-29	2	REPÚBLICA	No	Yes
6	733688164476661	5630279	F	23	Youth	2016-04-27 15:05:12.000	2016-04-29	2	GOIABEIRAS	No	No
7	344983394123	5630575	F	39	Adults	2016-04-27 15:39:58.000	2016-04-29	2	GOIABEIRAS	No	No
8	56394729949972	5638447	F	21	Youth	2016-04-29 08:02:16.000	2016-04-29	0	ANDORINHAS	No	No
9	78124564369297	5629123	F	19	Youth	2016-04-27 12:48:25.000	2016-04-29	2	CONQUISTA	No	No

PART 14:

Connect Power BI Desktop to Snowflake and create a dashboard.

Dashboard title:

Healthcare Appointment No-Show Dashboard

Create these DAX measures.

Total Appointments

Missed Appointments

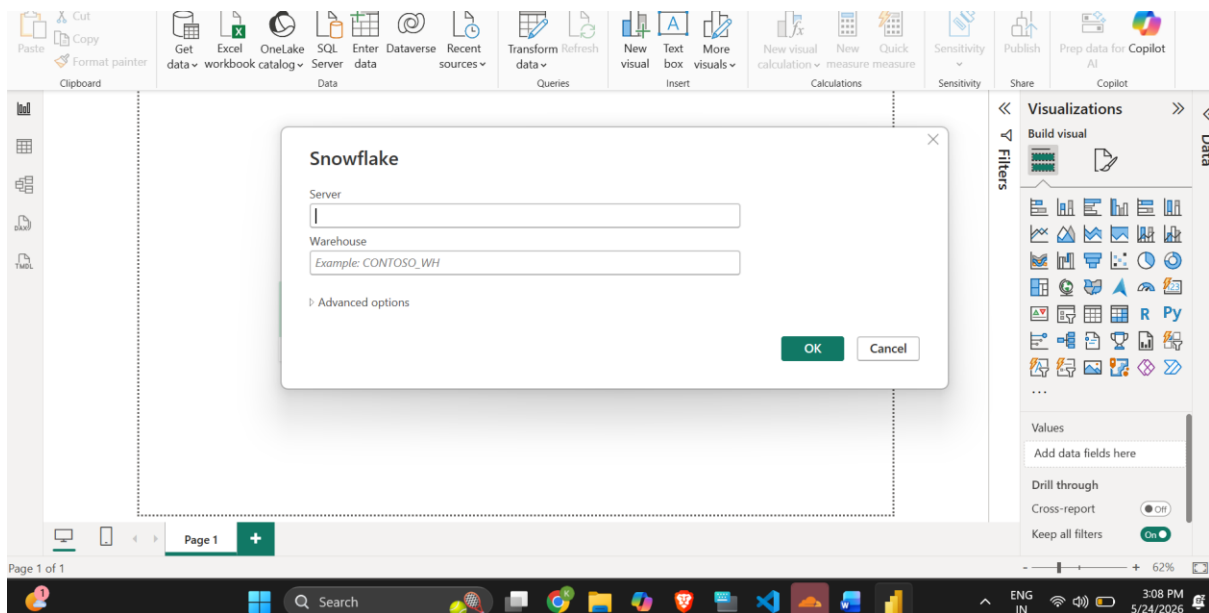
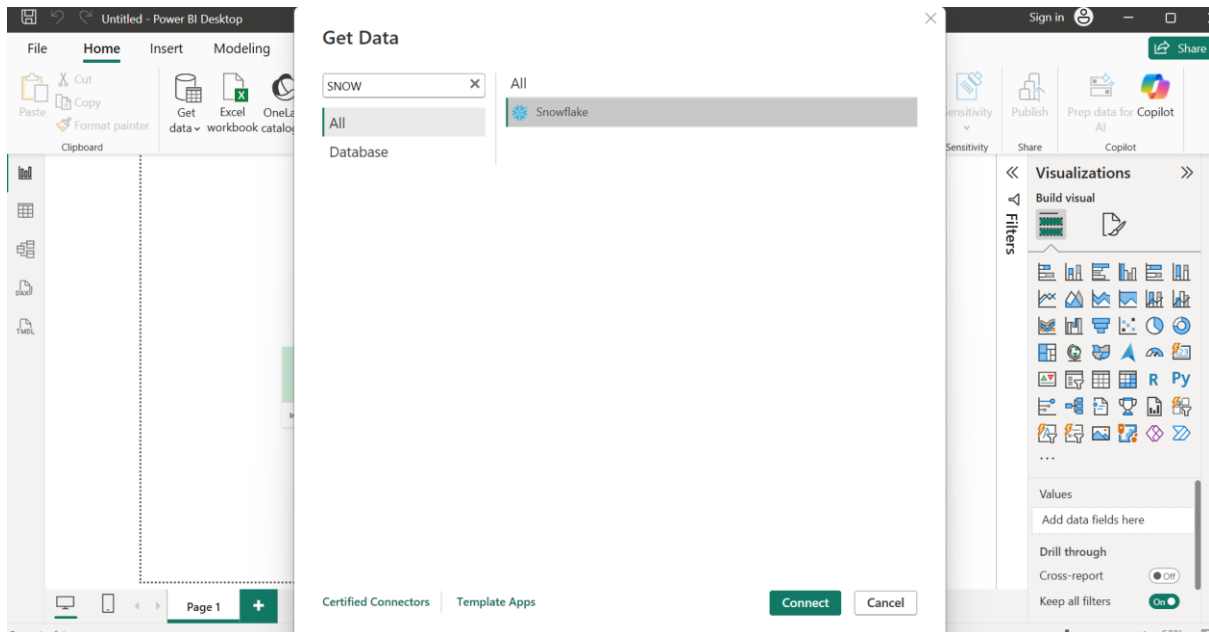
Visited Appointments

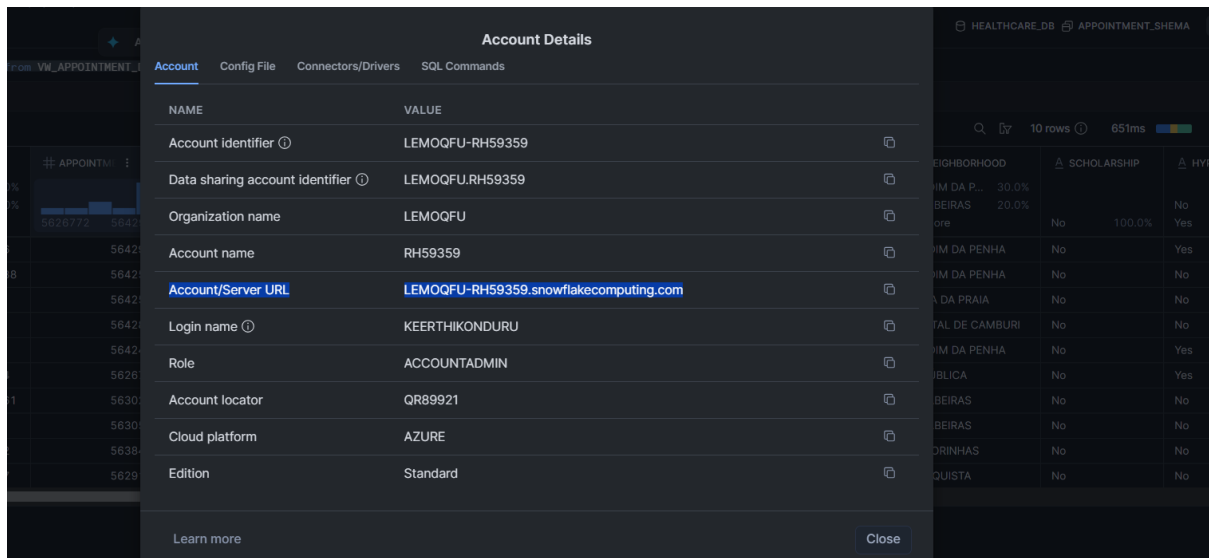
No Show Rate

Average Waiting Days

Attendance Status Distribution -> Donut Chart

No-Show by Age Group -> Clustered Column Chart





SERVER NAME IN POWER BI = SERVER URL IN SNOWFLAKE

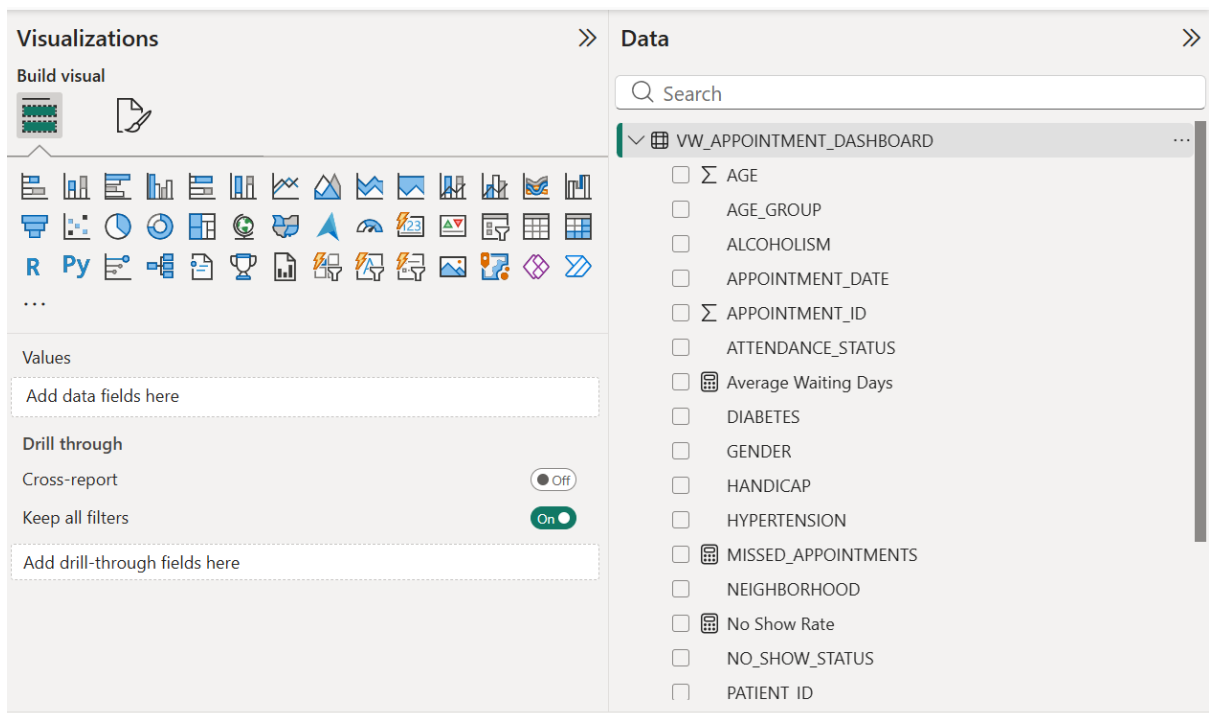
IN THE NEXT PAGE

GIVE USERNAME AND PASSWORD OF SNOWFLAKE IN THE POWERBI

WE CAN SEE DATABASES OF SNOWFLAKE

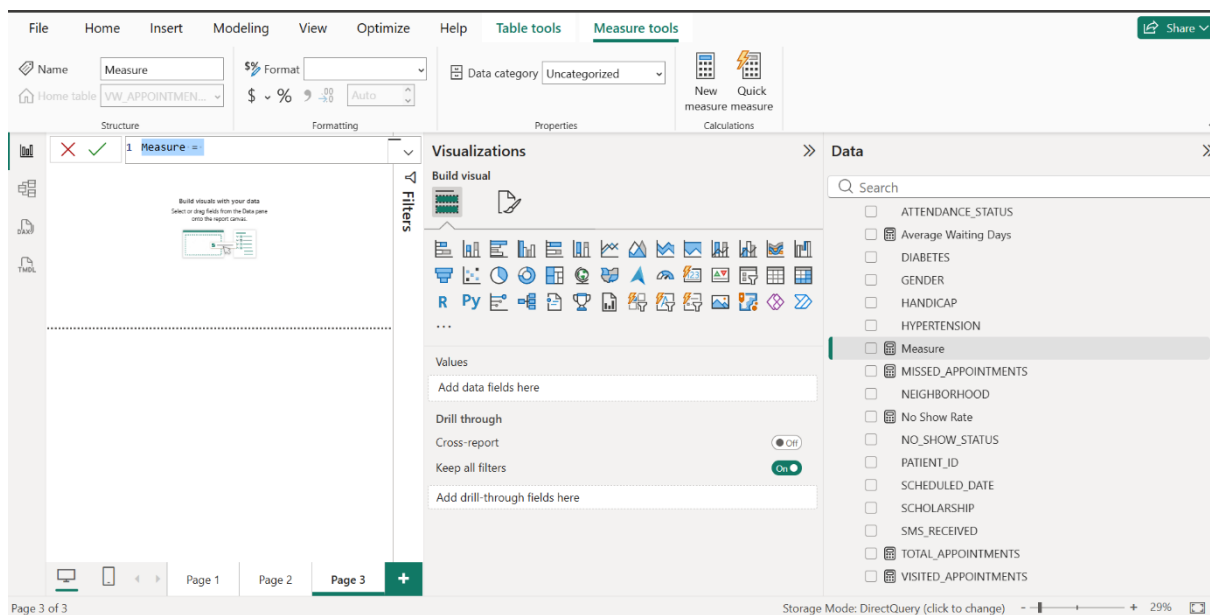
SELECT THE VIEW THAT WE CRAETED

SELECT DIRECT QUERY



HERE WE CAN SEE THE VIEW THAT WE SELECTED IN THE DATA

CREATING NEW DAX



Create these DAX measures.

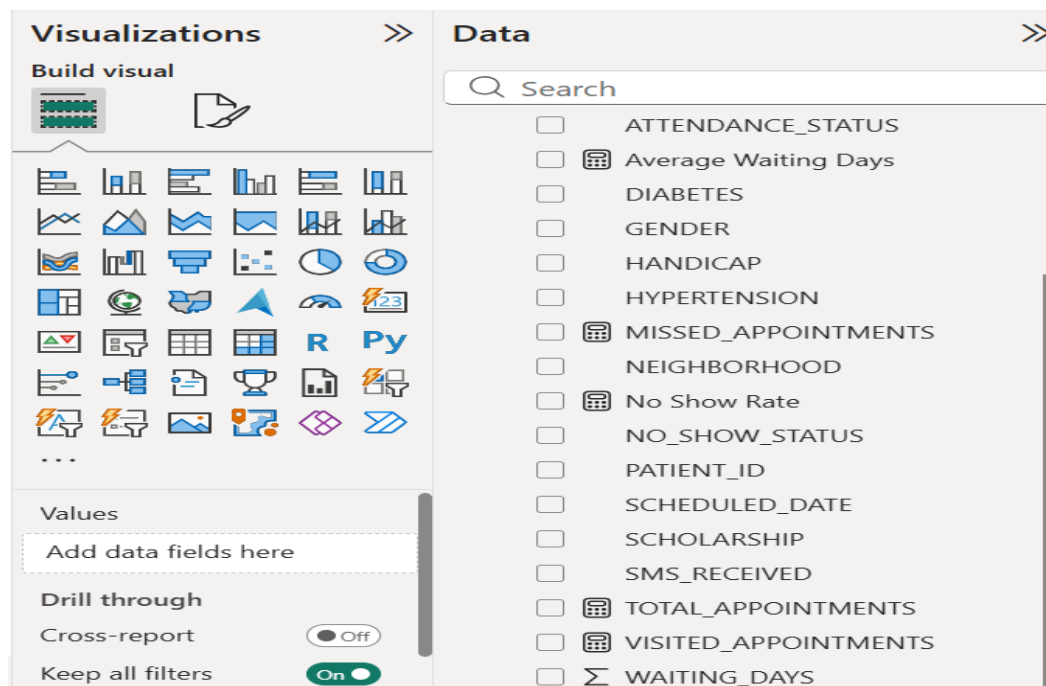
Total Appointments

Missed Appointments

Visited Appointments

No Show Rate

Average Waiting Days



THE PHONE SYMBOLS ARE THE MEASURE OR DAX THAT WE CREATED.

Dashboard title:

Healthcare Appointment No-Show Dashboard

Create these DAX measures.

Total Appointments

Missed Appointments

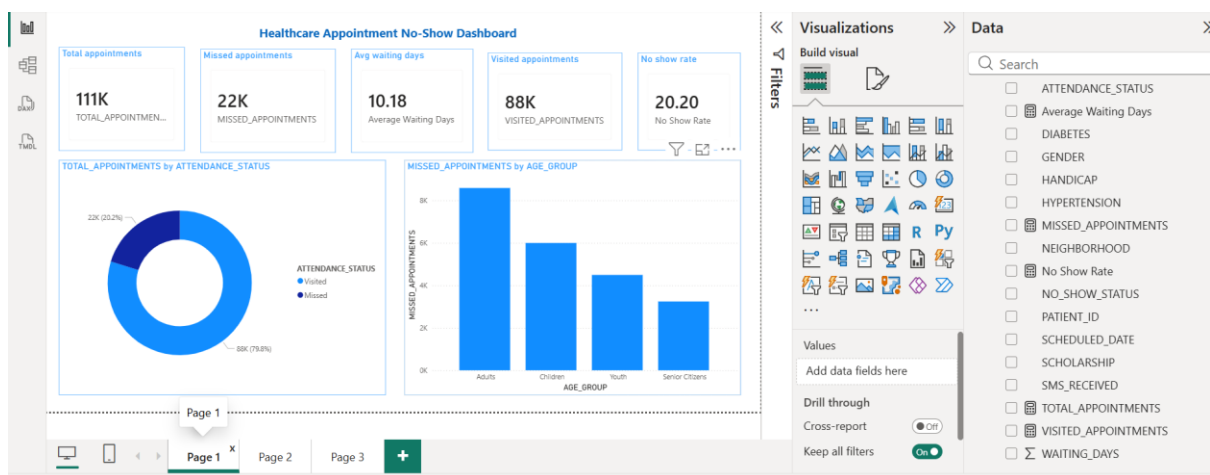
Visited Appointments

No Show Rate

Average Waiting Days

Attendance Status Distribution -> Donut Chart

No-Show by Age Group -> Clustered Column Chart



Does SMS reminder reduce no-show?

MISSED APPOINTMENTS AFTER SMS RECEIVED?

Which neighborhood has more missed appointments?

